

Mixed problems (3/3)

10.7–10.9

Outline

Antitrust split

Songs on CDs

Books on shelves

Antitrust split

Introduction: Antitrust split

- ▶ A company must split into **two companies**: every branch goes to one side or to the other
- ▶ The objective is **min-max**: we look at the product on which the two companies are most unbalanced
- ▶ We will see how an **absolute value** is written with two inequalities and a single auxiliary variable
- ▶ Above all we will see a case in which the LP relaxation **says nothing** ($z(LP) = 0$) and the bound must be looked for in integrality

Problem 10.7: antitrust split (1/2)

- ▶ A company has $s \in \mathbb{Z}_{\geq 1}$ branches and sells $r \in \mathbb{Z}_{\geq 1}$ products
- ▶ For every branch $i \in \{1, 2, \dots, s\}$ and every product $j \in \{1, 2, \dots, r\}$, the value $v_{ij} \in \mathbb{Q}_{\geq 0}$ is the revenue in millions of euros generated by that branch with that product
- ▶ Because of a new antitrust regulation the company must split into **two** smaller companies

Problem 10.7: antitrust split (2/2)

- ▶ Every branch is **indivisible** and must be assigned to exactly one of the two companies
- ▶ The company wants to partition the branches **minimising the maximum difference**, over all the products, between the revenues of the two new companies
- ▶ There is no constraint on the number of branches per company: the only link is the objective

The problem in words

- ▶ We **decide** which of the two companies each branch goes to
- ▶ **The objective:** to make the two companies as similar as possible, measuring similarity on the product where they do worst
- ▶ **The constraint:** every branch to a single company
- ▶ It is a **balanced partition** problem with several criteria, one per product

Data (model input)

Symbol	Type	Meaning
s	$\in \mathbb{Z}_{\geq 1}$	number of branches, index i
r	$\in \mathbb{Z}_{\geq 1}$	number of products, index j
v_{ij}	$\in \mathbb{Q}_{\geq 0}$	revenue of branch i with product j
T_j	$= \sum_{i=1}^s v_{ij}$	total revenue on product j

The datum T_j is not an independent input: it is computed from the v_{ij} and is needed to write the difference between the two companies.

Decision variables

One family of s binary variables is enough:

$$x_i = \begin{cases} 1 & \text{if branch } i \text{ goes to company } A \\ 0 & \text{if it goes to company } B \end{cases} \quad \forall i \in \{1, 2, \dots, s\}$$

plus a **free** variable z for the min-max.

- ▶ One could also declare $z \geq 0$: the constraints keep it above an absolute value, so the solution would not change
- ▶ It is left free because that is the choice one sees in the **dual**: the column of a free variable gives an *equality* constraint

One family or two?

- ▶ The classic statement introduces *two* families, x_i and y_i , with the constraint $x_i + y_i = 1$
- ▶ The two writings are equivalent, because $y_i = 1 - x_i$: substituting one obtains the same model with s **fewer** variables and s fewer constraints
- ▶ It is better to keep the aggregated form as long as the companies are *two*; with three or more companies the disaggregated form $\sum_k x_{ik} = 1$ is the only possible one, and it is the one that extends
- ▶ Recognising when two models are the same thing is part of the craft: here the check is a substitution

From revenue to difference

Let $T_j = \sum_{i=1}^s v_{ij}$ be the total revenue on product j .

- ▶ The revenue of company A on product j is $\sum_{i=1}^s v_{ij} x_i$
- ▶ That of company B is what is left, $T_j - \sum_{i=1}^s v_{ij} x_i$

Their difference is therefore

$$2 \sum_{i=1}^s v_{ij} x_i - T_j$$

- ▶ It is a signed quantity: the imbalance of product j is its **absolute value**

Model 10.7

Model 10.7 — antitrust split

$$\min z \quad (10.13a)$$

$$\text{subject to } z - 2 \sum_{i=1}^s v_{ij} x_i + T_j \geq 0, \quad \forall j \in \{1, 2, \dots, r\}, \quad (10.13b)$$

$$z + 2 \sum_{i=1}^s v_{ij} x_i - T_j \geq 0, \quad \forall j \in \{1, 2, \dots, r\}, \quad (10.13c)$$

$$x_i \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, s\}, \quad (10.13d)$$

$$z \geq 0. \quad (10.13e)$$

Model description

- ▶ The **objective function** is the worst imbalance, measured by the single variable z
- ▶ The constraints of the **first family**, one per product, say that z is no smaller than the imbalance of product j
- ▶ The constraints of the **second family**, again one per product, say that z is no smaller than its opposite
- ▶ Together they impose $z \geq |2 \sum_i v_{ij} x_i - T_j|$, and the minimisation turns them into an equality on the worst product

Absolute value and min-max in two lines

The two inequalities say

$$-z \leq 2 \sum_{i=1}^s v_{ij} x_i - T_j \leq z$$

that is $z \geq |2 \sum_{i=1}^s v_{ij} x_i - T_j|$ for every product j .

- ▶ Since z appears alone in the minimisation objective, at the optimum it equals exactly the **maximum of the absolute values**
- ▶ It is the min-max technique applied to the absolute value technique
- ▶ The direction “ z is no larger than necessary” is not imposed by any constraint, but by **optimality**

The model in gurobipy

```
m = gp.Model("antitrust")
x = m.addVars(s, vtype=GRB.BINARY, name="x")
z = m.addVar(lb=-GRB.INFINITY, name="z")
m.setObjective(z, GRB.MINIMIZE)
for j in range(r):
    tot = sum(v[i][j] for i in range(s))
    m.addConstr(z - 2 * gp.quicksum(
        v[i][j] * x[i] for i in range(s))
        + tot >= 0)
    m.addConstr(z + 2 * gp.quicksum(
        v[i][j] * x[i] for i in range(s))
        - tot >= 0)
```

The instance

$s = 4$ branches, $r = 3$ products:

v_{ij}	$j = 1$	$j = 2$	$j = 3$	branch total
$i = 1$	3	3	2	8
$i = 2$	6	8	5	19
$i = 3$	3	4	4	11
$i = 4$	2	7	9	18
T_j	14	22	20	

The totals per product are 14, 22 and 20; the overall revenue is 56.

Model 10.7 on the instance

The instance model

$$\begin{array}{ll}\min & z \\ \text{subject to} & -6x_1 - 12x_2 - 6x_3 - 4x_4 + z \geq -14 \\ & 6x_1 + 12x_2 + 6x_3 + 4x_4 + z \geq 14 \\ & -6x_1 - 16x_2 - 8x_3 - 14x_4 + z \geq -22 \\ & 6x_1 + 16x_2 + 8x_3 + 14x_4 + z \geq 22 \\ & -4x_1 - 10x_2 - 8x_3 - 18x_4 + z \geq -20 \\ & 4x_1 + 10x_2 + 8x_3 + 18x_4 + z \geq 20 \\ & x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\} \\ & z \geq 0\end{array}$$

Constructive heuristic: the primal bound (1/2)

- ▶ The branches are assigned in **decreasing order of total revenue**
- ▶ Each one goes to the company that currently has the smaller total
- ▶ It is the analogue of the LPT rule for scheduling
- ▶ It is a constructive heuristic: a single solution is built, one element at a time, never backtracking

Constructive heuristic: the primal bound (2/2)

On the instance the branch totals are 8, 19, 11, 18, so the order is 2, 4, 3, 1:

- ▶ branch 2 to A ($A = 19$), branch 4 to B ($B = 18$)
- ▶ branch 3 to B ($B = 29$), branch 1 to A ($A = 27$)

One gets $A = \{1, 2\}$ and $B = \{3, 4\}$, with differences 4, 0 and 6 on the three products:

$$z(MILP) \leq UB = 6$$

The worst product is the third one, and it gives the value of the heuristic solution.

Dual of the LP relaxation of model 10.7

We associate $\lambda_j \geq 0$ with the constraints of the first family and $\mu_j \geq 0$ with those of the second one.

Dual of the LP relaxation of model 10.7

$$\max \sum_{j=1}^r T_j (\mu_j - \lambda_j) \quad (10.14a)$$

$$\text{subject to } \sum_{j=1}^r (\lambda_j + \mu_j) = 1, \quad (10.14b)$$

$$2 \sum_{j=1}^r v_{ij} (\mu_j - \lambda_j) \leq 0, \quad \forall i \in \{1, 2, \dots, s\}, \quad (10.14c)$$

$$\lambda_j \geq 0, \quad \forall j \in \{1, 2, \dots, r\}, \quad (10.14d)$$

$$\mu_j \geq 0, \quad \forall j \in \{1, 2, \dots, r\}. \quad (10.14e)$$

Dual on the instance

The instance model

$$\begin{aligned} \max & -14\lambda_1 - 22\lambda_2 - 20\lambda_3 + 14\mu_1 + 22\mu_2 + 20\mu_3 \\ \text{subject to} & \quad \lambda_1 + \lambda_2 + \lambda_3 + \mu_1 + \mu_2 + \mu_3 = 1 \\ & -6\lambda_1 - 6\lambda_2 - 4\lambda_3 + 6\mu_1 + 6\mu_2 + 4\mu_3 \leq 0 \\ & -12\lambda_1 - 16\lambda_2 - 10\lambda_3 + 12\mu_1 + 16\mu_2 + 10\mu_3 \leq 0 \\ & -6\lambda_1 - 8\lambda_2 - 8\lambda_3 + 6\mu_1 + 8\mu_2 + 8\mu_3 \leq 0 \\ & -4\lambda_1 - 14\lambda_2 - 18\lambda_3 + 4\mu_1 + 14\mu_2 + 18\mu_3 \leq 0 \\ & \lambda_1, \quad \lambda_2, \quad \lambda_3 \geq 0 \\ & \mu_1, \quad \mu_2, \quad \mu_3 \geq 0 \end{aligned}$$

Dual description

- ▶ λ_j and μ_j are the **prices** of the two constraints that bound the imbalance of product j , one from above and one from below
- ▶ The **objective** prices the total T_j of each product at those values
- ▶ The **column of z** : the variable appears in every constraint with coefficient 1 and in the objective with cost 1, so the prices of the $2r$ constraints must sum to one — the dual distributes a single unit of weight among the products
- ▶ The **columns of the x_i** : moving branch i changes the imbalance of product j by $2v_{ij}$, and the weighted sum of these moves cannot be positive

A dual solution by hand

Take

$$\bar{\lambda}_1 = \bar{\mu}_1 = \frac{1}{2}, \quad \text{everything else} = 0$$

- ▶ The column of z is satisfied: $\frac{1}{2} + \frac{1}{2} = 1$
- ▶ We have $\mu_j - \lambda_j = 0$ for every product, so the columns of the x_i are satisfied as well
- ▶ The value is

$$LB = 14 \cdot \left(\frac{1}{2} - \frac{1}{2} \right) = 0$$

Here the dual cannot do better (1/2)

Observation 1: Here the dual cannot do better

No feasible dual solution is worth more than zero. Setting $\theta_j = \mu_j - \lambda_j$, the objective is $\sum_{j=1}^r T_j \theta_j$ and the columns of the x_i impose $\sum_{j=1}^r v_{ij} \theta_j \leq 0$ for every branch; summing over all the branches

$$\sum_{j=1}^r T_j \theta_j \leq 0.$$

The dual is therefore worth at most 0 and, by strong duality, $z(LP) = 0$.

Here the dual cannot do better (2/2)

Observation 2: The primal certificate is even simpler

The fractional solution

$$x_i = \frac{1}{2} \quad \forall i, \quad z = 0$$

*is feasible for the relaxation and balances every product exactly.
Half a branch to each company: legitimate for the LP, meaningless for the problem.*

A combinatorial bound, product by product (1/2)

If the relaxation says nothing, one must look elsewhere. For every product j one computes, looking at *that product alone*, the smallest imbalance achievable:

$$g_j = \min_{S \subseteq \{1,2,\dots,s\}} \left| 2 \sum_{i \in S} v_{ij} - T_j \right|$$

- ▶ It is the classic partition problem on a single column, solved by enumeration (2^s subsets)
- ▶ Every feasible partition is in particular a partition for product j , hence

$$z(MILP) \geq \max_j g_j$$

A combinatorial bound, product by product (2/2)

Product	total T_j	g_j
1	14	2
2	22	0
3	20	2

On product 1 the values are 3, 6, 3, 2: the sum 7 is not reachable by any subset, and the best is 6 against 8, that is $g_1 = 2$.

$$z(MILP) \geq LB = 2$$

It is a bound that the LP relaxation cannot see, because it comes from **integrality** and not from the constraints.

Optimal solution

The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there.

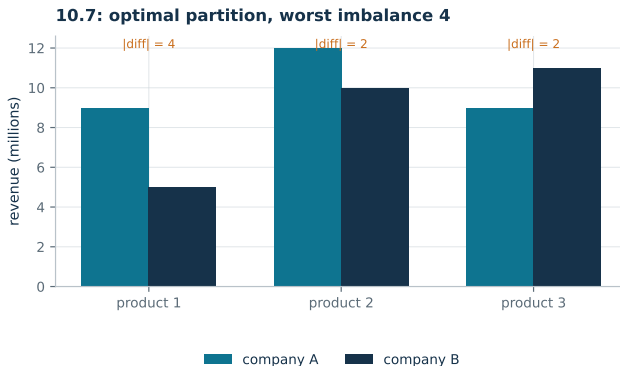
Company	branches	revenues per product
<i>A</i>	2, 3	9, 12, 9
<i>B</i>	1, 4	5, 10, 11

The differences are 4, 2 and 2, so $z(MILP) = 4$ and the chain of bounds is

$$2 = LB \leq z(MILP) = 4 \leq UB = 6, \quad z(LP) = z(LP^+) = 0$$

The heuristic gap is 50%.

The revenue of the two optimal companies



Product by product, the two bars are almost equal: the highlighted difference is the imbalance, and the largest of the three is $z(MILP) = 4$.

Additional remarks

- ▶ With a single product the problem is *number partitioning*, one of Karp's twenty-one NP-complete problems; with several products it is its vector version, and it stays NP-hard
- ▶ The combinatorial bound can be **strengthened**: instead of the maximum of the g_j one can look, for every pair of products, for the minimum of the maximum of the two imbalances — the same idea as *surrogate relaxation*
- ▶ The min-max objective is not the only possible one: minimising the **sum** of the differences is another legitimate choice, and gives a model with r auxiliary variables instead of one

Modelling question: min-sum instead of min-max

One wants to minimise the *sum* of the differences over all the products instead of the worst difference.

How does the model change? Is the optimal partition the same?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
UB	6	heuristic solution
LB	2	certified bound
$z(MILP)$	4	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 50.0\%$
- ▶ Certified gap, known *before* solving the MILP: $UB - LB = 4$,
that is $\frac{UB - LB}{|UB|} = 66.7\%$

Worked variant: two inseparable branches (1/3)

Branches 1 and 2 share their premises and must stay in the same company. One adds the single linear constraint

$$x_1 - x_2 = 0$$

that is “both to A or both to B ”.

- ▶ The only feasible partitions with 1 and 2 together are $\{1, 2\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$, with their complements
- ▶ The best one is exactly the one found by the heuristic: the optimum worsens from 4 to 6

Worked variant: two inseparable branches (2/3)

- ▶ The equality adds to the dual a **free** variable σ , with a plus sign in the column of branch 1 and a minus in that of branch 2
- ▶ Its right-hand side is zero, so the dual objective does not change: the relaxation is worth 0 as in the base problem, and for the same reason
- ▶ The useful bound is **combinatorial** again: keeping branches 1 and 2 together, the best possible imbalance on the three products is 4, 0 and 2, so $z(MILP) \geq 4$
- ▶ With the branches free it was 2: **tying two branches raises the bound**

Worked variant: two inseparable branches (3/3)

	value	what it is
UB	6	heuristic solution
LB	4	smallest imbalance with branches 1 and 2 tied
$z(LP)$	0	relaxation without the bounds
$z(LP^+)$	0	relaxation with the bounds
$z(MILP)$	6	MILP optimum

Here the heuristic is optimal, and the certified gap closes to $UB - LB = 2$.

Summary: Antitrust split

- ▶ A single family of binaries x_i and one free variable z are enough: the absolute value is written with two inequalities per product
- ▶ The dual distributes a **single unit of weight** among the products, because z is free and appears everywhere with coefficient 1
- ▶ The LP relaxation is worth 0 and is useless: the solution $x_i = 1/2$ balances everything
- ▶ The useful bound is combinatorial, product by product:
 $2 = LB \leq z(MILP) = 4 \leq UB = 6$, with a heuristic gap of 50%

Songs on CDs

Introduction: Songs on CDs

- ▶ A collection of songs must be spread over several CDs, keeping the CDs as **homogeneous** as possible in duration
- ▶ The objective is a **difference**: the duration of the longest CD minus that of the shortest one
- ▶ We need **two** auxiliary variables, pushed in opposite directions by the objective
- ▶ We will see a two-line **parity** argument that settles the problem before even calling the solver

Problem 10.8: songs on CDs (1/2)

- ▶ A music production company publishes a collection of $n \in \mathbb{Z}_{\geq 1}$ songs
- ▶ For every song $i \in \{1, 2, \dots, n\}$, the value $d_i \in \mathbb{Q}_{\geq 1}$ is the duration in minutes
- ▶ The collection must be distributed over $m \in \mathbb{Z}_{\geq 1}$ compact discs
- ▶ Every song must be assigned to **exactly one** CD

Problem 10.8: songs on CDs (2/2)

- ▶ Every CD $j \in \{1, 2, \dots, m\}$ must contain at least $w_j \in \mathbb{Z}_{\geq 1}$ songs
- ▶ The company wants to **minimise the difference** between the total duration of the longest CD and that of the shortest one
- ▶ There is no cap on the duration of a CD: the only criterion is the balance between the media

The problem in words

- ▶ We **decide** which CD every song ends up on
- ▶ **The objective**: CDs as homogeneous as possible in duration
- ▶ **The constraints**: every song on one CD only, and no CD too sparse
- ▶ It is a **balancing** problem, a close relative of scheduling on parallel machines

Data (model input)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of songs, index i
m	$\in \mathbb{Z}_{\geq 1}$	number of CDs, index j
d_i	$\in \mathbb{Q}_{\geq 1}$	duration in minutes of song i
w_j	$\in \mathbb{Z}_{\geq 1}$	minimum number of songs on CD j

The total duration $\sum_{i=1}^n d_i$ is not an input, but it is the number the bound will come from.

Decision variables

$n m$ binary variables:

$$x_{ij} = \begin{cases} 1 & \text{if song } i \text{ goes on CD } j \\ 0 & \text{otherwise} \end{cases}$$

and two non-negative continuous variables:

y = duration of the longest CD, z = duration of the shortest CD

- The x_{ij} are the real decisions; y and z only serve to write the objective

Model 10.8

Model 10.8 — songs on CDs

$$\min \quad y - z \quad (10.15a)$$

$$\text{subject to} \quad \sum_{j=1}^m x_{ij} = 1, \quad \forall i \in \{1, 2, \dots, n\}, \quad (10.15b)$$

$$\sum_{i=1}^n x_{ij} \geq w_j, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.15c)$$

$$-\sum_{i=1}^n d_i x_{ij} + y \geq 0, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.15d)$$

$$\sum_{i=1}^n d_i x_{ij} - z \geq 0, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.15e)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, n\}, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.15f)$$

$$y, z \geq 0. \quad (10.15g)$$

Model description

- ▶ The **objective function** is the difference between the longest and the shortest duration
- ▶ The **assignment** constraints, one per song, say that every song ends up on exactly one CD (n equalities)
- ▶ The **minimum filling** constraints, one per CD, impose the minimum number of songs
- ▶ The **maximum** constraints push y above every duration, the **minimum** ones push z below every duration

Two auxiliary variables, two opposite directions

- ▶ The maximum constraints say $y \geq \sum_{i=1}^n d_i x_{ij}$ for every CD: y is *at least* the maximum
- ▶ The minimum constraints say $z \leq \sum_{i=1}^n d_i x_{ij}$ for every CD: z is *at most* the minimum

The objective $\min(y - z)$ pushes y down and z up, so at the optimum

$$y = \max_j \sum_{i=1}^n d_i x_{ij}, \quad z = \min_j \sum_{i=1}^n d_i x_{ij}$$

- ▶ One needs an objective that pushes *both* variables in the right direction, otherwise one of the two would stay undetermined

The model in gurobipy (1/2)

```
mod = gp.Model("cds")
x = mod.addVars(n, m, vtype=GRB.BINARY,
                name="x")
y = mod.addVar(name="y")
z = mod.addVar(name="z")
mod.setObjective(y - z, GRB.MINIMIZE)
```



The model in gurobipy (2/2)

```
mod.addConstrs((x.sum(i, "*") == 1
                 for i in range(n)), name="song")
mod.addConstrs((x.sum("*, j) >= w[j]
                 for j in range(m)), name="minimum")
mod.addConstrs((y - gp.quicksum(
    d[i] * x[i, j] for i in range(n)) >= 0
    for j in range(m)), name="maximum")
mod.addConstrs((gp.quicksum(
    d[i] * x[i, j] for i in range(n)) - z >= 0
    for j in range(m)), name="min_duration")
```

The instance

$n = 6$ songs, $m = 2$ CDs, $w_j = 1$ for both:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$
d_i	5	6	7	3	4	10

The total duration is 35 minutes: an **odd** number, and it will come in handy.

Model 10.8 on the instance

The instance model

$$\begin{array}{ll}
 \min & y - z \\
 \text{subject to} & x_{11} + x_{12} = 1 \\
 & x_{21} + x_{22} = 1 \\
 & x_{31} + x_{32} = 1 \\
 & x_{41} + x_{42} = 1 \\
 & x_{51} + x_{52} = 1 \\
 & x_{61} + x_{62} = 1 \\
 & x_{11} + x_{21} + x_{31} + x_{41} + x_{51} + x_{61} \geq 1 \\
 & x_{12} + x_{22} + x_{32} + x_{42} + x_{52} + x_{62} \geq 1 \\
 & -5x_{11} - 6x_{21} - 7x_{31} - 3x_{41} - 4x_{51} - 10x_{61} + y \geq 0 \\
 & -5x_{12} - 6x_{22} - 7x_{32} - 3x_{42} - 4x_{52} - 10x_{62} + y \geq 0 \\
 & 5x_{11} + 6x_{21} + 7x_{31} + 3x_{41} + 4x_{51} + 10x_{61} - z \geq 0 \\
 & 5x_{12} + 6x_{22} + 7x_{32} + 3x_{42} + 4x_{52} + 10x_{62} - z \geq 0 \\
 & x_{11}, x_{12}, x_{21}, x_{22}, x_{31}, x_{32}, x_{41}, x_{42}, x_{51}, x_{52}, x_{61}, x_{62} \in \{0, 1\} \\
 & y \geq 0 \\
 & z \geq 0
 \end{array}$$

Two heuristics compared: the primal bound (1/2)

The insertion rule is always the same — every song on the **currently shortest CD** — but the order in which the songs are scanned changes the result a lot.

- ▶ It is a constructive heuristic: a single solution is built, one element at a time, never backtracking
- ▶ **(a) Natural order:** $5 \rightarrow \text{CD1}$, $6 \rightarrow \text{CD2}$, $7 \rightarrow \text{CD1}$, $3 \rightarrow \text{CD2}$, $4 \rightarrow \text{CD2}$, $10 \rightarrow \text{CD1}$
- ▶ Final durations 22 and 13: difference 9

Two heuristics compared: the primal bound (2/2)

- ▶ **(b) Decreasing duration (LPT):** $10 \rightarrow \text{CD1}$, $7 \rightarrow \text{CD2}$, $6 \rightarrow \text{CD2}$, $5 \rightarrow \text{CD1}$, $4 \rightarrow \text{CD2}$, $3 \rightarrow \text{CD1}$
- ▶ Final durations 18 and 17: difference 1

The same rule, with two different orders, gives 9 and 1. The better one is kept:

$$z(MILP) \leq UB = 1$$

The order in which the elements are scanned is part of the heuristic, not a detail.

Dual of the LP relaxation of model 10.8

α_i free for the assignment, $\beta_j \geq 0$ for the minimum filling, $\gamma_j \geq 0$ and $\delta_j \geq 0$ for the two durations.

Dual of the LP relaxation of model 10.8

$$\max \sum_{i=1}^n \alpha_i + \sum_{j=1}^m w_j \beta_j \quad (10.16a)$$

$$\text{subject to } \sum_{j=1}^m \gamma_j = 1, \quad (10.16b)$$

$$\sum_{j=1}^m \delta_j = 1, \quad (10.16c)$$

$$\alpha_i + \beta_j - d_i \gamma_j + d_i \delta_j \leq 0, \quad \forall i \in \{1, 2, \dots, n\}, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.16d)$$

$$\alpha_i \geq 0, \quad \forall i \in \{1, 2, \dots, n\}, \quad (10.16e)$$

$$\beta_j \geq 0, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.16f)$$

$$\gamma_j \geq 0, \quad \forall j \in \{1, 2, \dots, m\}, \quad (10.16g)$$

$$\delta_j \geq 0, \quad \forall j \in \{1, 2, \dots, m\}. \quad (10.16h)$$

Dual on the instance

The instance model

$$\begin{aligned}
 & \max \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6 + \beta_1 + \beta_2 \\
 & \text{subject to} \\
 & \qquad \qquad \qquad \gamma_1 \quad + \gamma_2 \qquad \qquad = 1 \\
 & \qquad \qquad \qquad \qquad \delta_1 \quad + \delta_2 = 1 \\
 & \alpha_1 \qquad \qquad + \beta_1 \quad - 5\gamma_1 \quad + 5\delta_1 \leq 0 \\
 & \alpha_1 \qquad \qquad + \beta_2 \quad - 5\gamma_2 \quad + 5\delta_2 \leq 0 \\
 & \alpha_2 \qquad \qquad + \beta_1 \quad - 6\gamma_1 \quad + 6\delta_1 \leq 0 \\
 & \alpha_2 \qquad \qquad + \beta_2 \quad - 6\gamma_2 \quad + 6\delta_2 \leq 0 \\
 & \alpha_3 \qquad \qquad + \beta_1 \quad - 7\gamma_1 \quad + 7\delta_1 \leq 0 \\
 & \alpha_3 \qquad \qquad + \beta_2 \quad - 7\gamma_2 \quad + 7\delta_2 \leq 0 \\
 & \alpha_4 \qquad \qquad + \beta_1 \quad - 3\gamma_1 \quad + 3\delta_1 \leq 0 \\
 & \alpha_4 \qquad \qquad + \beta_2 \quad - 3\gamma_2 \quad + 3\delta_2 \leq 0 \\
 & \alpha_5 \qquad \qquad + \beta_1 \quad - 4\gamma_1 \quad + 4\delta_1 \leq 0 \\
 & \alpha_5 \qquad \qquad + \beta_2 \quad - 4\gamma_2 \quad + 4\delta_2 \leq 0 \\
 & \alpha_6 + \beta_1 \quad - 10\gamma_1 \quad + 10\delta_1 \leq 0 \\
 & \alpha_6 + \beta_2 \quad - 10\gamma_2 \quad + 10\delta_2 \leq 0 \\
 & \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6 \qquad \qquad \qquad \geq 0 \\
 & \qquad \qquad \beta_1, \beta_2 \qquad \qquad \qquad \geq 0 \\
 & \qquad \qquad \qquad \gamma_1, \gamma_2 \qquad \qquad \geq 0 \\
 & \qquad \qquad \qquad \delta_1, \delta_2 \geq 0
 \end{aligned}$$

Dual description (1/2)

- ▶ α_i is the **value of song** i : it is free because the assignment constraint is an equality
- ▶ β_j is the **price** of the minimum number of songs on CD j
- ▶ γ_j and δ_j are the **weights** with which CD j enters the maximum and the minimum duration respectively
- ▶ The **objective** prices the songs and the thresholds w_j at these values

Dual description (2/2)

- ▶ **Columns of y and z :** each of the two variables appears in one constraint per CD and in the primal objective with coefficient ± 1 , so the respective weights must sum to one
- ▶ **Columns of the x_{ij} :** putting song i on CD j satisfies its assignment constraint (α_i), contributes to the minimum number of songs (β_j) and moves both durations by d_i
- ▶ The balance cannot be positive, because the x_{ij} do not appear in the objective

A dual solution by hand

Let us set

$$\bar{\gamma}_j = \bar{\delta}_j = \frac{1}{m}, \quad \bar{\alpha} = \bar{\beta} = 0$$

- ▶ The two equality constraints are satisfied: m weights equal to $1/m$ sum to one
- ▶ The columns of the x_{ij} become

$$-\frac{d_i}{m} + \frac{d_i}{m} = 0 \leq 0$$

- ▶ The value is $LB = 0$: the choice that favours no CD is worth nothing

Here too the relaxation is mute

Observation 3: Here too the relaxation is mute

That the relaxation is worth zero is best seen from the primal side: putting $x_{ij} = 1/m$ for every song and every CD, every CD “lasts”

$$\frac{35}{2} = 17.5 \text{ minutes}$$

and the difference is zero. It is feasible for the relaxation and meaningless for the problem, because a song cannot be split. The LP relaxation of balancing models is almost always like this: it levels everything and is worth zero.

A parity argument that settles the problem

The durations are **integers** and the CDs are two: the durations q_1 and q_2 are integers adding up to $q = 35$, which is *odd*.

- ▶ Two integers adding up to an odd number cannot be equal
- ▶ Their difference $|q_1 - q_2|$ has the same parity as q , that is it is odd
- ▶ An odd non-negative difference is at least 1

$$z(MILP) \geq LB = 1$$

The LPT heuristic reaches exactly 1: the two bounds coincide and the heuristic solution is **optimal**, and one knows it *before* calling the solver.

Optimal solution

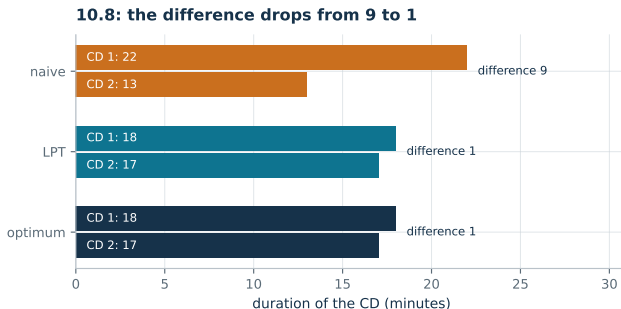
The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there.

	songs	duration
CD 1	1, 2, 3	18
CD 2	4, 5, 6	17

$$1 = LB \leq z(MILP) = 1 \leq UB = 1, \quad z(LP) = z(LP^+) = 0$$

The heuristic gap is zero. The solver's solution is not the LPT one (songs 6, 1, 4 and 3, 2, 5), but it has the same value: the problem has **several optima**.

The duration of the two CDs in the three solutions



From left to right: naive heuristic (22 against 13), LPT (18 against 17) and the optimum, which matches LPT.

Additional remarks

- ▶ The parity argument holds as written for $m = 2$. With $m = 3$ it must be redone: 35 is not divisible by 3, so the three durations cannot all be equal, but the minimum spread is no longer immediate (on the instance the optimum is 2, with durations 13, 11, 11)
- ▶ The constraint $\sum_{i=1}^n x_{ij} \geq w_j$ with $w_j = 1$ never bites at the optimum (an empty CD would give $z = 0$), but it is needed in the model, because without it the relaxation would have degenerate solutions
- ▶ Minimising $y - z$ is not the same as minimising y (the makespan): it is the difference between **balancing** and **finishing early**

Modelling question: a smaller medium

CD 1 is a reduced medium and cannot exceed 15 minutes.

How does the model change? What is the new optimum?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
UB	1	heuristic solution
LB	1	certified bound
$z(MILP)$	1	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 0.0\%$
- ▶ Certified gap, known *before* solving the MILP: $UB - LB = 0$,
that is $\frac{UB - LB}{|UB|} = 0.0\%$

It is the clearest case of the course: the MILP confirms, it does not discover.

Worked variant: three CDs (1/3)

The collection is distributed over **three** CDs instead of two: only the datum m changes, from 2 to 3 (and the list w).

- ▶ On the instance the optimum rises to 2: with three CDs the total duration 35 does not split into equal parts
- ▶ The 10-minute song is the structural constraint: if it sits alone on a CD, the other two share 25 and the maximum is at least 13
- ▶ If it goes with the 3-minute song, one gets durations 13, 11, 11 and the difference is 2

Worked variant: three CDs (2/3)

- ▶ The structure of the dual does not change: the columns of y and z ask the γ and the δ to sum to one, and spreading them evenly, $\gamma_j = \delta_j = 1/3$, is the choice that favours no CD
- ▶ The dual is worth 0, like the relaxation: a fractional solution splits every song into three equal parts and levels the durations
- ▶ The useful bound comes from **integrality** again: the durations are integers summing to 35, which is not a multiple of 3, so they cannot all be equal and the difference is at least 1
- ▶ The LPT heuristic leaves 2, which is the optimum: the certified gap stays open by one unit

Worked variant: three CDs (3/3)

	value	what it is
UB	2	heuristic solution
LB	1	parity of the durations on three CDs
$z(LP)$	0	relaxation without the bounds
$z(LP^+)$	0	relaxation with the bounds
$z(MILP)$	2	MILP optimum

With two CDs the parity bound settled the problem; with three CDs it leaves one unit of uncertainty.

Summary: Songs on CDs

- ▶ Two auxiliary variables, y for the maximum and z for the minimum, pushed in opposite directions by the objective $\min(y - z)$
- ▶ The same heuristic with two different orders gives 9 and 1: the order is part of the algorithm
- ▶ The LP relaxation is worth 0, as in all balancing models: $x_{ij} = 1/m$ levels everything
- ▶ The **parity** argument gives $LB = 1 = UB = z(MILP)$: the problem is settled before solving it

Books on shelves

Introduction: Books on shelves

- ▶ An **assignment problem with capacity**: every book goes on one shelf, and every shelf has a maximum width
- ▶ What is new is the cost: the height of a shelf is that of the **tallest book** placed on it
- ▶ We will see the **maximum variable** in its disaggregated form, one constraint per book and no big-M
- ▶ We will also see a first-fit heuristic that **can fail**, and a combinatorial bound stronger than the dual one

Problem 10.9: books on shelves (1/2)

- ▶ A library needs to arrange $n \in \mathbb{Z}_{\geq 1}$ books on shelves
- ▶ Every book $b \in \{1, 2, \dots, n\}$ has
 - ▶ width $w_b \in \mathbb{Z}_{\geq 1}$
 - ▶ height $h_b \in \mathbb{Z}_{\geq 1}$
- ▶ There are $m \in \mathbb{Z}_{\geq 1}$ shelves available, each of maximum width $c \in \mathbb{Q}_{>0}$

Problem 10.9: books on shelves (2/2)

- ▶ Every book must be assigned to **exactly one** shelf
- ▶ The total width of the books on a shelf cannot exceed c
- ▶ The **height** of a shelf is that of the tallest book placed on it
- ▶ The library wants to **minimise the sum of the heights** of the shelves

The problem in words

- ▶ We **decide** which shelf every book goes on
- ▶ **The objective:** minimum sum of the heights (it is the timber that is saved)
- ▶ **The constraints:**
 - ▶ every book on one shelf only
 - ▶ no shelf wider than c
- ▶ It is a variant of **bin packing** in which the cost of a container depends on its contents

Data (model input)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of books, $b \in \{1, 2, \dots, n\}$
m	$\in \mathbb{Z}_{\geq 1}$	number of shelves, $s \in \{1, 2, \dots, m\}$
w_b	$\in \mathbb{Z}_{\geq 1}$	width of book b
h_b	$\in \mathbb{Z}_{\geq 1}$	height of book b
c	$\in \mathbb{Q}_{>0}$	maximum width of a shelf

Decision variables

$n \times m$ binary variables:

$$x_{bs} = \begin{cases} 1 & \text{if book } b \text{ goes on shelf } s, \\ 0 & \text{otherwise} \end{cases}$$

m non-negative continuous variables:

$$y_s = \text{height of shelf } s, \quad \forall s \in \{1, 2, \dots, m\}$$

Model 10.9

Model 10.9 — books on shelves

$$\min \sum_{s=1}^m y_s \quad (10.17a)$$

$$\text{subject to } \sum_{s=1}^m x_{bs} = 1, \quad \forall b \in \{1, 2, \dots, n\}, \quad (10.17b)$$

$$\sum_{b=1}^n w_b x_{bs} \leq c, \quad \forall s \in \{1, 2, \dots, m\}, \quad (10.17c)$$

$$-h_b x_{bs} + y_s \geq 0, \quad \forall b \in \{1, 2, \dots, n\}, \forall s \in \{1, 2, \dots, m\}, \quad (10.17d)$$

$$x_{bs} \in \{0, 1\}, \quad \forall b \in \{1, 2, \dots, n\}, \forall s \in \{1, 2, \dots, m\}, \quad (10.17e)$$

$$y_s \geq 0, \quad \forall s \in \{1, 2, \dots, m\}. \quad (10.17f)$$

Model description

- ▶ The **objective function** is the total height of the shelves
- ▶ The **assignment** constraints, one per book (n equalities), say that every book sits on exactly one shelf
- ▶ The **width** constraints, one per shelf (m constraints), are the available capacity
- ▶ The **height** constraints, one per book–shelf pair ($n m$ constraints), push y_s above the height of every book placed on that shelf

The link between the variables

The height constraints read

$$y_s \geq h_b x_{bs}$$

- ▶ If book b is on shelf s then $y_s \geq h_b$
- ▶ Otherwise the constraint is empty: it only says $y_s \geq 0$
- ▶ Since y_s appears in the minimisation objective with a positive coefficient, at the optimum it equals exactly the **largest** height among the books present, and it is 0 if the shelf is empty
- ▶ It is technique 6.5, the **maximum variable**, in its **disaggregated** form: one constraint per book instead of a single one with a big-M

An empty shelf costs nothing

- ▶ With $y_s \geq 0$ and no other constraint, an empty shelf has $y_s = 0$ and does not contribute to the objective
- ▶ It follows that giving the library **more shelves can never worsen** the optimum
- ▶ But it need not improve it: splitting the books over more shelves means paying more heights
- ▶ On the instance, with three shelves instead of two, the optimum stays 15

The model in gurobipy (1/2)

```
mod = gp.Model("shelves")
x = mod.addVars(n, m, vtype=GRB.BINARY, name="x")
y = mod.addVars(m, name="y")
mod.setObjective(y.sum(), GRB.MINIMIZE)
mod.addConstrs((x.sum(b, "*") == 1
                 for b in range(n)), name="book")
```

The model in gurobipy (2/2)

```
mod.addConstrs((gp.quicksum(w[b] * x[b, s]
                             for b in range(n)) <= c
                for s in range(m)), name="width")
mod.addConstrs((y[s] - h[b] * x[b, s] >= 0
                for b in range(n)
                for s in range(m)), name="height")
```

The instance

$n = 4$ books, $m = 2$ shelves, $c = 10$:

	$b = 1$	$b = 2$	$b = 3$	$b = 4$
w_b	3	5	4	6
h_b	8	5	7	4

The total width of the books is 18, the overall capacity $2 \cdot 10 = 20$:
the margin is tight.

Model 10.9 on the instance

The instance model

$$\begin{array}{llllll} \min & & & & & y_1 + y_2 \\ \text{subject to} & x_{11} + x_{12} & & & & = 1 \\ & & x_{21} + x_{22} & & & = 1 \\ & & & x_{31} + x_{32} & & = 1 \\ & & & & x_{41} + x_{42} & = 1 \\ & 3x_{11} & + 5x_{21} & + 4x_{31} & + 6x_{41} & \leq 10 \\ & 3x_{12} & + 5x_{22} & + 4x_{32} & + 6x_{42} & \leq 10 \\ & -8x_{11} & & & & + y_1 \geq 0 \\ & -8x_{12} & & & & + y_2 \geq 0 \\ & & -5x_{21} & & & + y_1 \geq 0 \\ & & -5x_{22} & & & + y_2 \geq 0 \\ & & & -7x_{31} & & + y_1 \geq 0 \\ & & & -7x_{32} & & + y_2 \geq 0 \\ & & & & -4x_{41} & + y_1 \geq 0 \\ & & & & -4x_{42} & + y_2 \geq 0 \\ & x_{11}, x_{12}, x_{21}, x_{22}, x_{31}, x_{32}, x_{41}, x_{42} & & & & \in \{0, 1\} \\ & & & & & y_1, y_2 \geq 0 \end{array}$$

Constructive heuristic: the primal bound (1/2)

The rule is **first-fit**: every book on the first shelf where it fits.

- ▶ It is a constructive heuristic: a single solution is built, one element at a time, never backtracking
- ▶ The order in which the books are considered changes everything

(a) Decreasing height (1, 3, 2, 4)

- ▶ book 1 (width 3) on shelf 1; book 3 (width 4) on shelf 1, which reaches 7; book 2 (width 5) on shelf 2
- ▶ book 4, of width 6, is left: shelf 1 has 3 of residual width, shelf 2 has 5
- ▶ **The heuristic fails**

Constructive heuristic: the primal bound (2/2)

(b) Decreasing width (4, 2, 3, 1)

- ▶ book 4 (width 6) on shelf 1
- ▶ book 2 (width 5) on shelf 2
- ▶ book 3 (width 4) on shelf 1, which reaches 10
- ▶ book 1 (width 3) on shelf 2, which reaches 8

Heights: $\max(4, 7) = 7$ and $\max(5, 8) = 8$, sum 15.

$$z(MILP) \leq UB = 15$$

The right order depends on the constraint, not on the objective

Observation 4: The right order depends on the constraint, not on the objective

The decreasing-height order is the one suggested by the objective, but the constraint that can make an insertion infeasible is the width: that is what one must sort on. It is the same logic as first-fit decreasing for bin packing, where one sorts by size and not by value.

A constructive heuristic that may fail is not wrong in itself — one only has to foresee the case and change the order — but it must be said explicitly, because a heuristic that sometimes returns nothing provides no bound at all.

Dual of the LP relaxation of model 10.9

One **free** α_b per assignment, one $\beta_s \leq 0$ per width constraint, one $\gamma_{bs} \geq 0$ per height constraint.

Dual of the LP relaxation of model 10.9

$$\max \quad \sum_{b=1}^n \alpha_b + c \sum_{s=1}^m \beta_s \quad (10.18a)$$

$$\text{subject to} \quad \sum_{b=1}^n \gamma_{bs} \leq 1, \quad \forall s \in \{1, 2, \dots, m\}, \quad (10.18b)$$

$$\alpha_b + w_b \beta_s - h_b \gamma_{bs} \leq 0, \quad \forall b \in \{1, 2, \dots, n\}, \quad \forall s \in \{1, 2, \dots, m\}, \quad (10.18c)$$

$$\alpha_b \geq 0, \quad \forall b \in \{1, 2, \dots, n\}, \quad (10.18d)$$

$$\beta_s \leq 0, \quad \forall s \in \{1, 2, \dots, m\}, \quad (10.18e)$$

$$\gamma_{bs} \geq 0, \quad \forall b \in \{1, 2, \dots, n\}, \quad \forall s \in \{1, 2, \dots, m\}. \quad (10.18f)$$

Dual on the instance

The instance model

$$\begin{array}{llllllll}
 \max & \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + 10\beta_1 + 10\beta_2 & & & & & & \\
 \text{subject to} & & \gamma_{11} & +\gamma_{21} & +\gamma_{31} & +\gamma_{41} & & \leq 1 \\
 & & & \gamma_{12} & +\gamma_{22} & +\gamma_{32} & +\gamma_{42} & \leq 1 \\
 \alpha_1 & & +3\beta_1 & -8\gamma_{11} & & & & \leq 0 \\
 \alpha_1 & & & +3\beta_2 & -8\gamma_{12} & & & \leq 0 \\
 \alpha_2 & & +5\beta_1 & & -5\gamma_{21} & & & \leq 0 \\
 \alpha_2 & & & +5\beta_2 & & -5\gamma_{22} & & \leq 0 \\
 \alpha_3 & +4\beta_1 & & & & -7\gamma_{31} & & \leq 0 \\
 \alpha_3 & & +4\beta_2 & & & & -7\gamma_{32} & \leq 0 \\
 \alpha_4 & +6\beta_1 & & & & & -4\gamma_{41} & \leq 0 \\
 \alpha_4 & & +6\beta_2 & & & & & -4\gamma_{42} \leq 0 \\
 \alpha_1, \alpha_2, \alpha_3, \alpha_4 & & & & & & & \geq 0 \\
 & & \beta_1, \beta_2 & & & & & \geq 0 \\
 & & & \gamma_{11}, \gamma_{12}, \gamma_{21}, \gamma_{22}, \gamma_{31}, \gamma_{32}, \gamma_{41}, \gamma_{42} & & & & \geq 0
 \end{array}$$

Dual description

- ▶ α_b is the **value of book b** ; β_s the (non-positive) price of the width of shelf s ; γ_{bs} the price of the link “the height of shelf s covers book b ”
- ▶ **Columns of the y_s :** the height of shelf s costs 1 in the primal objective, and the prices of the links that push it up cannot be worth more
- ▶ **Columns of the x_{bs} :** putting book b on shelf s satisfies its assignment constraint (α_b), takes up w_b of width and forces the height up to h_b ; the balance cannot be positive, because the x_{bs} do not appear in the objective

A dual solution by hand

- ▶ Set $\bar{\beta} = 0$: the width is not priced
- ▶ Concentrate all the γ weight on the tallest book, number 1 with $h_1 = 8$:

$$\bar{\gamma}_{1s} = 1 \quad \forall s, \quad \bar{\gamma}_{bs} = 0 \quad \text{for } b \neq 1$$

- ▶ The constraints on the columns of the y_s become $1 \leq 1$
- ▶ Those on the columns of the x_{bs} become $\alpha_1 \leq 8$ and $\alpha_b \leq 0$ for $b \neq 1$: take $\bar{\alpha}_1 = 8$ and $\bar{\alpha}_b = 0$

$$LB = 8$$

It is the obvious remark: the shelf holding the tallest book is at least as tall as that book.

A stronger combinatorial bound

The total width of the books is 18 and every shelf holds 10: at least

$$\left\lceil \frac{18}{10} \right\rceil = 2$$

shelves must be non-empty. One of them holds the tallest book and measures at least 8; the other one holds at least one book, so it measures at least $\min_{b \neq 1} h_b = 4$.

$$z(MILP) \geq LB = 8 + 4 = 12$$

- ▶ Better than the dual bound 8: the jump comes from **integrality**
- ▶ The relaxation can put half a book on each shelf and pay half the height twice, that is 8 again

Optimal solution

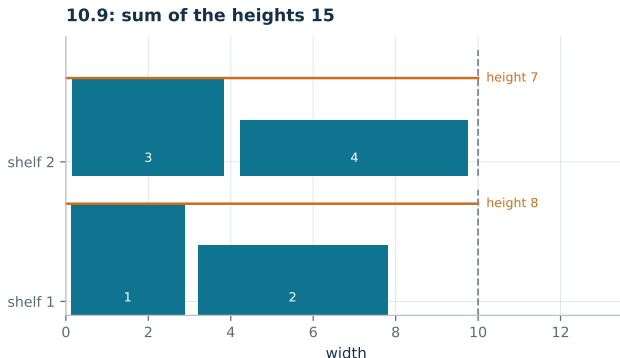
The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there.

	books	width	height
shelf 1	1, 2	8 out of 10	8
shelf 2	3, 4	10 out of 10	7

$$12 = LB \leq z(MILP) = 15 \leq UB = 15, \quad z(LP) = z(LP^+) = 8$$

The heuristic gap is **zero**; the certified gap, before solving the MILP, is $(15 - 12)/15 = 20\%$.

The optimal arrangement



The books on each shelf, with the shelf height marked in orange and the maximum width dashed.

Additional remarks (1/2)

- ▶ The problem is a variant of **bin packing** in which the cost of a container is not fixed but depends on its contents
- ▶ It is known as *bin packing with item fragmentation* or, in the warehousing literature, as *shelf space allocation*
- ▶ The model has an annoying **symmetry**: swapping the two shelves gives the same solution with the same value
- ▶ On larger instances it is worth breaking it, for instance by imposing $y_1 \geq y_2 \geq \dots \geq y_m$

Additional remarks (2/2)

- ▶ The **disaggregated** form of the height constraints uses $n m$ constraints
- ▶ The **aggregated** form with a big-M,

$$y_s \geq h_b - M(1 - x_{bs})$$

would use just as many and is **weaker**

- ▶ Here the disaggregated one costs nothing and is therefore preferable

Modelling question: a third shelf

The library buys a third shelf, as wide as the others.

What is the new optimum?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
UB	15	heuristic solution
LB	12	certified bound
$z(MILP)$	15	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 0.0\%$
- ▶ Certified gap, known *before* solving the MILP (it cannot have $z(MILP)$ in the denominator): $UB - LB = 3$, that is $\frac{UB - LB}{|UB|} = 20.0\%$

Worked variant: wider shelves (1/3)

The shelves are 12 wide instead of 10: only the datum c changes.

- ▶ The arrangement $\{1, 2, 3\}$ on the first shelf becomes feasible, of width $3 + 5 + 4 = 12$ and height $\max(8, 5, 7) = 8$
- ▶ On the second shelf only book $\{4\}$ is left, of width 6 and height 4
- ▶ The sum is $8 + 4 = 12$: the optimum drops from 15 to 12

Worked variant: wider shelves (2/3)

- ▶ The **dual bound** stays 8: the width does not enter the certificate, because the recipe puts the whole γ weight on the tallest book, number 1, 8 tall
- ▶ The **combinatorial** argument is still needed: the total width is 18 and one shelf holds 12, so the non-empty shelves are still at least two, one at least 8 tall and the other at least 4, that is $LB = 12$
- ▶ Here the combinatorial bound is **exact**: the three shorter books sit on the same shelf as the tallest one, and the fourth stays alone
- ▶ What changes is the heuristic: first-fit by decreasing width fits the four books on two shelves 5 and 8 tall, and drops from 15 to 13

Worked variant: wider shelves (3/3)

	value	what it is
UB	13	heuristic solution
LB	12	shelves used and minimum heights
$z(LP)$	8	relaxation without the bounds
$z(LP^+)$	8	relaxation with the bounds
$z(MILP)$	12	optimum of the MILP

The certified bound 12 matches the optimum: the remaining gap is entirely the heuristic's.

Summary: Books on shelves

- ▶ The link $y_s \geq h_b x_{bs}$ is the **maximum variable** in disaggregated form: $n m$ constraints and no big-M
- ▶ In first-fit the order must be chosen on the **constraint** (the width), not on the objective (the height): by decreasing height the heuristic fails
- ▶ Bounds on the instance: $LB = 8$ from the dual, $LB = 12$ from the combinatorial argument, $z(LP) = z(LP^+) = 8$, $z(MILP) = 15$, $UB = 15$
- ▶ Heuristic gap 0%, certified gap 20%; with shelves 12 wide one gets $UB = 13$, $LB = 12$ and $z(MILP) = 12$

Final summary (1/3)

Selection with alternative modes (10.1, 10.2):

- ▶ one family of binaries per mode and a mutual exclusion constraint $\sum_m x_i^{(m)} \leq 1$
- ▶ the sum of the binaries of an object is the indicator “taken, one way or the other”

Counts with a minimum lot (10.3):

- ▶ integer quantity variables with a binary switching them on
- ▶ the semi-integer link $q_j y_j \leq x_j \leq M_j y_j$: zero or else at least the threshold

Final summary (2/3)

Covering with containers (10.4, 10.5):

- ▶ packs of fixed composition are bought: integer variables, not binary ones
- ▶ the excess is either paid for or wasted, and this changes the objective function

Splitting and balancing (10.6–10.9):

- ▶ a set divided among containers, with the quality measured by the imbalance between containers
- ▶ min–max, difference between the largest and the smallest, sum of the maxima: all are linearised with an auxiliary variable and one family of constraints

Final summary (3/3)

When the LP relaxation is weak:

- ▶ a fractional solution can split every object in half and level the containers: the bound collapses, sometimes down to zero
- ▶ the useful bound must be looked for in **combinatorial** arguments: parity of a count, minimum number of containers, dominance of one class
- ▶ these arguments come from integrality and the dual of the relaxation cannot see them

The method stays the same:

- ▶ constructive heuristic for UB , dual solution by hand for LB , solver for $z(MILP)$
- ▶ the certified gap is what is known *before* solving the MILP